Adaptive CASTNet: Traffic-State-Dependent Graph Sampling Attention for Multi-Horizon Traffic Flow Forecasting

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Abstract

Correct prediction of the traffic flow is conditioned by representation of temporal dynamics and spatial connections between sensors on roads. In this work, we analyze an adaptive version of CASTNet algorithm, which reweights a pre-defined road graph based on the current traffic state encoding and performs multi-head graph attention sampling top-k. It is tested on the PEMS04 and PEMS08 data sets with 15-, 30-, and 60-minutes horizon. Besides, the proposed approach is compared with traditional LSTM, GCN-LSTM, the spatiotemporal Transformers-based baseline, and static CASTNet-Lite. Moreover, speed forecasting is analyzed separately in METR-LA data set. The LSTM is the most accurate method on the two data sets in seed-42 PEMS experiments by overall MAE and RMSE. Adaptive CASTNet does not perform better than the basic CASTNet-Lite on average in PEMS experiments and has additional computational complexity. Thus, the current experiment represents a very carefully analyzed negative result – traffic-state-dependent graph reweighing is not providing an advantage.

Keywords: Traffic-flow forecasting Spatio-temporal learning Graph neural networks Graph attention Adaptive adjacency

1 Introduction

Traffic flow forecasting is an important part of intelligent transportation systems. Traffic management, route guidance, signal coordination and operational planning can be based on reliable short- and medium-term forecasts. This task is challenging due to

the temporal dependence and spatial coupling of measurements from road sensors: a change at one detector may propagate to nearby locations, while the strength of these relationships can vary with the current traffic state.

Graph neural networks provide a natural representation of sensor networks, while temporal convolution, recurrent networks, and attention mechanisms model changes over time. CASTNet combines graph convolution, graph-sampling attention, and temporal feature extraction to balance local structure and longer-range dependencies [**?**]. Hierarchical spatio-temporal graph convolution and Transformer approaches further illustrate the value of combining graph-based spatial representations with attention-based temporal modelling [**?**]. Earlier approaches include diffusion convolutional recurrent networks [**?**], spatio-temporal graph convolutional networks [**?**], Graph WaveNet [**?**], adaptive graph convolutional recurrent networks [**?**], and spatial-temporal synchronous graph convolution [**?**].

A common modelling choice is to use a fixed road graph throughout the sequence. A fixed graph is interpretable and preserves road-network structure, but it may not fully represent traffic-state-dependent relationships. This work investigates whether a lightweight adaptive graph learner can reweight the known road graph using the latest encoded traffic state. The key research question is: *Does adapting sensor-to-sensor edge weights to the current traffic state improve multi-horizon traffic-flow forecasting over a static CASTNet-inspired graph-attention model?*

The proposed Adaptive CASTNet is an experimental extension, not a claim of reproducing the original CASTNet architecture. It retains static road graph as a structural prior, estimates non-negative pairwise similarity between current sensor representations, maintains a limited set of strongest neighbors and applies sampled multi-head graph attention. We compare this model with 4 baselines on PEMS04 and PEMS08 through experiments. Another METR-LA experiment is a traffic-speed forecasting task, which is reported separately since its target variable is different from the PEMS flow target.

Contributions are: (1) The Traffic State Conditioned Graph Reweight Module that operates under the constraint of the pre-defined road graph. (2) Top-k neighbors selection in implementation to restrict the attention neighborhood. (3) Controlled experiments with LSTM, GCN-LSTM, Hierarchical Spatio-Temporal Transformer, and CASTNet-Lite without traffic information. (4) Multi-horizon, low flow, and inference time analysis. Notably, the experiments presented have been conducted with one random seed only and therefore represent preliminary results that cannot be treated as multi-seed statistical findings.

2 Related Work and Critical Synthesis

2.1 Graph-based traffic forecasting

DCRNN models traffic dynamics using bidirectional random walks on a directed graph and an encoder–decoder recurrent architecture [**?**]. STGCN uses graph convolution and temporal convolution in a fully convolutional framework [**?**]. Graph WaveNet combines adaptive dependency learning with dilated temporal convolutions [**?**]. AGCRN learns node-specific patterns and an adaptive graph within a recurrent graph-convolution

framework [**?**]. STSGCN explicitly models localized synchronous spatial-temporal correlations [**?**]. These methods demonstrate that the road graph and temporal signal should be considered jointly, but they differ in how the graph is defined, updated, and used.

2.2 Graph sampling and attention

The first reference in this study is CASTNet: Convolution Augmented Graph Sampling Transformer Network for Traffic Flow Forecasting [**?**]. CASTNet uses graph convolution for local structure learning and graph sampling attention for modeling the global dependencies along with temporal feature learning. The reasoning behind graph sampling in CASTNet is applicable to the large-scale sensor networks since paying attention to every possible combination of sensors may be costly. The current architecture draws inspiration from CASTNet as it employs local graph convolution, temporal convolution, top-k graph sampling attention, and an additional graph learner. This network is not a clone of CASTNet, hence the name CASTNet-Lite.

The second reference is the hierarchical spatio-temporal graph convolutional and Transformer network for traffic-flow forecasting [**?**]. The relevance of this model is that it combines the spatial graph encoding with temporal encoding via a Transformer. For this project, a baseline hierarchical-STTransformer model has been designed by implementing graph convolution, temporal convolution, and a two-layer Transformer for the temporal sequence of each individual sensor. It should be noted that the model implemented is a baseline project based on the modeling approach of the paper.

2.3 Research gap and positioning

Spatial structure, temporal modeling, adaptive relationships, and attention have been established by the literature to be useful. The specific question that will be answered in this paper is whether the re-weighting of a road graph, based on a state, adds any significant advantage when compared to a CASTNet-inspired sampled-attention model. In terms of comparison between CASTNet-Lite and Adaptive CASTNet, the difference can be more directly assessed compared to different architectures. There has been no positive difference shown between CASTNet-Lite and Adaptive CASTNet for the datasets of PEMS04 and PEMS08; the negative finding is relevant and has been discussed.

3 Datasets and Experimental Protocol

3.1 PEMS04 and PEMS08

The primary experiments use PEMS04 and PEMS08. PEMS04 contains 307 sensors and 16,992 five-minute time steps; PEMS08 contains 170 sensors and 17,856 five-minute time steps. The source tensors contain traffic-flow, speed, and occupancy channels. Only channel 0 (flow) is used as both input and prediction target in the primary experiments. The sensor distance CSV files contain `from`, `to`, and `cost` fields.

For each dataset, the time axis is split chronologically into 60% training, 20% validation, and 20% test segments. No global shuffle is applied before splitting. The

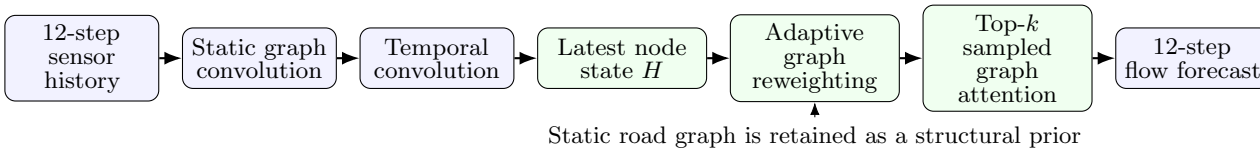


Fig. 1 Self-created schematic of the Adaptive CASTNet pipeline. The dynamic graph learner reweights, rather than replaces, the static road graph.

training mean and standard deviation are used to z-normalize the flow series; predictions are inverse-transformed before metrics are computed. Each sample uses the previous 12 time steps (60 minutes) to predict the next 12 time steps (60 minutes). The resulting window counts are 10,172/3,375/3,376 for PEMS04 and 10,690/3,548/3,549 for PEMS08 (train/validation/test).

3.2 Road graph construction

The graph is derived from the supplied sensor-distance CSV. For each finite distance d_{ij}, a Gaussian affinity is computed as $\exp[-(d_{ij}/\sigma)^2]$, where σ is the standard deviation of finite distances. The matrix is symmetrized using the elementwise maximum with its transpose, affinities below 0.05 are removed, and self-loops are added. Symmetric degree normalization gives $\hat{A} = D^{-1/2}AD^{-1/2}$. The resulting unnormalized graphs contain 859 non-zero entries for PEMS04 and 592 for PEMS08, including self-loops.

3.3 Supplementary METR-LA experiment

METR-LA is tested independently for the traffic-speed problem. This dataset is analyzed with 207 sensor readings and 34,272 time steps at five minutes' interval. There is 60/20/20 split by chronological order and both input and output are equal to 12 steps. This result will not be combined with the PEMS result since it is about the speed and not traffic flow.

4 Proposed Methodology

4.1 Overall architecture

Given a batch of historical traffic observations $X \in \mathbb{R}^{B\times T\times N\times 1}$ and normalized static adjacency $\hat{A} \in \mathbb{R}^{N\times N}$, a graph-convolution layer first aggregates neighboring sensor features. A temporal convolution then extracts local temporal patterns. The encoded representation at the latest input time is used as the current node state $H \in \mathbb{R}^{B\times N\times d}$. The proposed dynamic graph learner reweights the static graph from this state, after which sampled multi-head graph attention and a feed-forward refinement block produce the 12-step forecast.

4.2 Static spatial and temporal encoding

The graph-convolution layer applies normalized neighborhood aggregation followed by a learned linear projection and ReLU activation:

$$Z_t = \mathrm{ReLU}\Big(\hat{A}X_tW_g + b_g\Big). \tag{1}$$

A one-dimensional temporal convolution with kernel size 3 is applied independently to each sensor's hidden sequence. Its output is added residually to the graph-convolution representation and layer-normalized. This produces $Z \in \mathbb{R}^{B\times T\times N\times d}$; the latest state is $H = Z_{:,T,:,:}$.

4.3 Traffic-state-dependent graph learner

Two learned linear maps project each node state into query and key embeddings. After L2 normalization, pairwise similarity is computed as

$$S_{bij} = \mathrm{ReLU}\left(\frac{q(H_{bi})^\top k(H_{bj})}{\|q(H_{bi})\|_2\,\|k(H_{bj})\|_2}\right). \tag{2}$$

The static adjacency constrains the learned relationships:

$$\tilde{A}_{bij} = \hat{A}_{ij}(1 + S_{bij}). \tag{3}$$

For each source node, only the largest $k = 8$ values are retained; the retained weights are row-normalized to obtain A^{dyn}. This construction preserves the static road graph's support: a pair with zero static affinity cannot become a new edge solely because its learned feature similarity is high. The graph is sample-dependent because S is computed from the current batch's latest node states.

4.4 Sampled graph attention and forecasting head

Multi-head queries, keys, and values are computed for every node using an attention mechanism. Keys and values are gathered from the top-k nodes only, scaled dot-product attention is computed and the log of the selected edge weight is added to the attention weight. The output of the attention layer is passed through a projection layer and summed with the node state using a residual connection and then a LayerNorm layer. A two-layer feed-forward network with GELU non-linearity and dropout is used after another residual connection and LayerNorm layer. Linear layer converts the hidden node state into 12 future flow values.

This model employs hidden size 64, four attention heads, top-k of 8, and dropout of 0.10. The ratio of feed-forward expansion is four. In CASTNet-Lite ablation, we keep the same sampled attention mechanism but remove only the dynamic graph learner on top of the fixed normalized adjacency matrix.

4.5 Training procedure

Each of the five models is trained using AdamW (learning rate 10^{-3}, weight decay 10^{-4}), with batch size 16 and a maximum number of 50 epochs. The ReduceLROnPlateau schedule reduces the learning rate in case of 3 plateaus on validation-RMSE. Early stopping has patience 10, and the best checkpoint based on validation-RMSE is used. Gradient clipping is performed, and mixed precision is utilized in case of CUDA availability. The seed value is set to 42. Although there is a multiseed template in the notebook, the experiment was run with only one seed value (42).

5 Evaluation Metrics

The models are then tested based on MAE, RMSE, MAPE, and coefficient of determination (R^2) after inverse normalization. MAPE ignores those ground truth values that have an absolute value of 10-5 or less to avoid division by zero or near-zero values. The measurements are performed for all forecasting nodes and all horizons unless otherwise stated. The analysis of horizons gives results for 15, 30, and 60 minutes (3, 6, and 12 five-minute periods).

6 Results and Discussion

6.1 Overall PEMS results

Table **??** and Table **??** report the held-out test results from seed 42. On both datasets, the LSTM baseline achieves the lowest overall MAE and RMSE. Among the graph-attention models, CASTNet-Lite is slightly better than Adaptive CASTNet on the aggregate PEMS flow metrics. The dynamic graph learner therefore does not improve overall test performance in this run.

For PEMS04, the MAE and RMSE of the Adaptive CASTNet are 33.3694 and 51.2359, respectively, while that of the CASTNet-Lite are 33.1354 and 50.9036, respectively. For PEMS08, the MAE and RMSE of the Adaptive CASTNet are 35.8871 and 53.0729, respectively, while that of the CASTNet-Lite are 35.3573 and 52.9043, respectively. The Relative RMSE change recorded in the notebook is -0.65% for PEMS04 and -0.32% for PEMS08 in terms of improvement of the Adaptive CASTNet over the CASTNet-Lite model, which means a slight decrement rather than an increment.

6.2 Forecast horizon analysis

Table **??** presents the MAE values for the three forecast horizons. The error level rises as the forecast horizon increases for the graph attention network models. LSTM continues to be the best baseline model in both PEMS datasets. For the 15-minute forecast horizon, Adaptive CASTNet shows a slightly lower MAE than CASTNet-Lite in PEMS04 (26.6177 against 26.7082), but it performs poorly in the other two forecast horizons. In the case of PEMS08, CASTNet-Lite shows lower MAE values in all three forecast horizons.

Table 5 Parameter count and recorded inference time per batch.

Dataset / Model	Parameters	ms/batch
PEMS04 LSTM	334,948	0.3503
PEMS04 CASTNet-Lite	63,180	3.1351
PEMS04 Adaptive CASTNet	71,372	5.6371
PEMS08 LSTM	193,016	0.8072
PEMS08 CASTNet-Lite	63,180	1.8005
PEMS08 Adaptive CASTNet	71,372	2.6621

6.3 Low-flow subset analysis

A low-flow set is obtained by taking the 25th percentile of the flow distribution in the training set, and the threshold values for the PEMS04 and PEMS08 datasets are 78.0 and 110.0 respectively. It is important to note that this is an experimental definition for a low-flow regime and should not be confused for a general definition of traffic congestion. The test set size for the low-flow regime is 605 for the PEMS04 dataset and 659 for the PEMS08 dataset. For this set, LSTM achieves the least MAE and RMSE.

6.4 Model size and inference time

Adaptive CASTNet has 71,372 parameters that can be trained, while CASTNet-Lite has 63,180 parameters. In terms of the inference time per batch in a test dataset, the notebook shows that it takes Adaptive CASTNet 5.6371 ms on PEMS04 and 2.6621 ms on PEMS08, while CASTNet-Lite takes 3.1351 ms and 1.8005 ms respectively. The LSTM model is faster in the provided benchmark. Those metrics depend on the implementation and hardware used, and there is no sufficient metadata in the notebook to compare them across different devices.

6.5 Supplementary METR-LA speed results

The standalone experiment for METR-LA predicts the speed instead of traffic flow. For the 60-minute time step, CASTNet-Lite achieves MAE 7.8566 and RMSE 13.4523, while Adaptive CASTNet reaches MAE 7.9667 and RMSE 13.4740. In terms of those two evaluation metrics, the static model performs slightly better; Adaptive CASTNet has MAPE 16.71% against 16.47% and $R^2 = 0.6505$ against 0.6517 for CASTNet-Lite. This is a completely different task, and thus the numbers will not be taken into account when combining the results for the PEMS dataset. The additional experiment does not alter the key finding about the ineffectiveness of the adaptive graph module in current runs.

6.6 Interpretation and limitations

The conclusions derived from the findings are important despite the fact that the suggested approach does not improve on the baselines. First, since the LSTM method achieves a good performance, we conclude that the time-dependent signal is a very good predictor with the chosen data set and preprocessing scheme. Second, the performance

of the graph-based methods could be influenced by the way they are built, normalized and the distribution of the sensor-level flow values. Third, the adaptive graph is limited by the static graph and the similarity of only one hidden state of the last timestamp; this might be inadequate for capturing evolving relationships between the channels over longer times. Fourth, this research utilizes one seed, one input/forecast length and one target channel.

The following experiments will have to consider at least three seeds (42, 123, 2026), include mean and standard deviation, test out different k values for top-k, perform controlled ablation studies on the dynamic graph learner, include comparisons between a completely learned adjacency matrix and the static-support assumption. Information regarding the hardware used is to be included for run time comparisons. The result tables that are revised need to be generated through experiments again.

7 Conclusion and Future Work

An Adaptive CASTNet model was proposed in this paper for multi-step traffic flow prediction. The proposed framework utilizes static graph convolution, temporal convolution, graph reweighting based on traffic states, k-top sampled graph attention, and a residual feedforward forecasting head. Evaluation was performed on PEMS04 and PEMS08 datasets using LSTM, GCN-LSTM, a hierarchically stacked Transformer model, and CASTNet-Lite as baselines.

In the described seed-42 runs, the LSTM model produced the best results concerning MAE and RMSE for both PEMS sets. The proposed CASTNet with adaptation failed to show superiority over CASTNet-Lite in both aggregate PEMS measures and in the case of low flow, despite requiring extra computational resources. In the METR-LA speed study provided as additional material, the same modest performance gain was observed for CASTNet-Lite at the 60-minute horizon. Therefore, the key conclusion here is not about the usefulness of graph learning for predictions, but about the ineffectiveness of the examined one.

For future work, we plan to look into validating multiple seed approaches, more effective traffic state representations, time-series aggregation of estimated dynamic graphs, top-k and graph threshold sensitivity analysis, and testing the model over more datasets and forecast lengths. It can only be concluded that the model outperforms the baselines if supported by repeated experiments.